

Filter Summary Report: CG,TIA,Automated,simple,Z4,Z5

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Contents

1 Examined $H(z)$ for CG TIA Automated simple Z4 Z5: $\frac{Z_4 Z_5 g_m - Z_4}{2Z_4 g_m + 2Z_5 g_m + 2}$

$$H(z) = \frac{Z_4 Z_5 g_m - Z_4}{2Z_4 g_m + 2Z_5 g_m + 2}$$

2 AP

3 BP

3.1 BP-1 $Z(s) = \left(\infty, \infty, \infty, \frac{L_4 s}{C_4 L_4 s^2 + 1}, R_5, \infty \right)$

$$H(s) = \frac{s(L_4 R_5 g_m - L_4)}{2L_4 g_m s + 2R_5 g_m + s^2(2C_4 L_4 R_5 g_m + 2C_4 L_4) + 2}$$

Parameters:

Q: $\frac{\sqrt{C_4 R_5 g_m + \sqrt{C_4}}}{\sqrt{L_4 g_m}}$
 wo: $\frac{1}{\sqrt{C_4} \sqrt{L_4}}$
 bandwidth: $\frac{g_m}{\sqrt{C_4}(\sqrt{C_4 R_5 g_m + \sqrt{C_4}})}$
 K-LP: 0
 K-HP: 0
 K-BP: $\frac{R_5 g_m - 1}{2g_m}$
 Qz: None
 Wz: None

3.2 BP-2 $Z(s) = \left(\infty, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, R_5, \infty \right)$

$$H(s) = \frac{s(L_4 R_4 R_5 g_m - L_4 R_4)}{2R_4 R_5 g_m + 2R_4 + s^2(2C_4 L_4 R_4 R_5 g_m + 2C_4 L_4 R_4) + s(2L_4 R_4 g_m + 2L_4 R_5 g_m + 2L_4)}$$

Parameters:

Q: $\frac{\sqrt{C_4 R_4 R_5 g_m + \sqrt{C_4} R_4}}{\sqrt{L_4 R_4 g_m + \sqrt{L_4 R_5 g_m + \sqrt{L_4}}}}$
 wo: $\frac{1}{\sqrt{C_4} \sqrt{L_4}}$
 bandwidth: $\frac{\sqrt{L_4 R_4 g_m + \sqrt{L_4 R_5 g_m + \sqrt{L_4}}}}{\sqrt{C_4} \sqrt{L_4}(\sqrt{C_4 R_4 R_5 g_m + \sqrt{C_4} R_4})}$
 K-LP: 0
 K-HP: 0
 K-BP: $\frac{R_4 R_5 g_m - R_4}{2R_4 g_m + 2R_5 g_m + 2}$
 Qz: None
 Wz: None

4 BS

4.1 BS-1 $Z(s) = \left(\infty, \infty, \infty, L_4 s + \frac{1}{C_4 s}, R_5, \infty \right)$

$$H(s) = \frac{R_5 g_m + s^2(C_4 L_4 R_5 g_m - C_4 L_4) - 1}{2C_4 L_4 g_m s^2 + 2g_m + s(2C_4 R_5 g_m + 2C_4)}$$

Parameters:

Q: $\frac{\sqrt{L_4 g_m}}{\sqrt{C_4 R_5 g_m + \sqrt{C_4}}}$
 wo: $\frac{1}{\sqrt{C_4} \sqrt{L_4}}$
 bandwidth: $\frac{\sqrt{C_4 R_5 g_m + \sqrt{C_4}}}{\sqrt{C_4 L_4 g_m}}$
 K-LP: $\frac{R_5 g_m - 1}{2g_m}$

K-HP: $\frac{R_5 g_m - 1}{2g_m}$
K-BP: 0
Qz: None
Wz: $\frac{1}{\sqrt{C_4} \sqrt{L_4}}$

4.2 BS-2 $Z(s) = \left(\infty, \infty, \infty, \frac{R_4(C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, R_5, \infty \right)$

$$H(s) = \frac{R_4 R_5 g_m - R_4 + s^2 (C_4 L_4 R_4 R_5 g_m - C_4 L_4 R_4)}{2R_4 g_m + 2R_5 g_m + s^2 (2C_4 L_4 R_4 g_m + 2C_4 L_4 R_5 g_m + 2C_4 L_4) + s (2C_4 R_4 R_5 g_m + 2C_4 R_4) + 2}$$

Parameters:

Q: $\frac{\sqrt{L_4} R_4 g_m + \sqrt{L_4} R_5 g_m + \sqrt{L_4}}{\sqrt{C_4} R_4 R_5 g_m + \sqrt{C_4} R_4}$
wo: $\frac{1}{\sqrt{C_4} \sqrt{L_4}}$
bandwidth: $\frac{\sqrt{C_4} R_4 R_5 g_m + \sqrt{C_4} R_4}{\sqrt{C_4} \sqrt{L_4} (\sqrt{L_4} R_4 g_m + \sqrt{L_4} R_5 g_m + \sqrt{L_4})}$
K-LP: $\frac{R_4 R_5 g_m - R_4}{2R_4 g_m + 2R_5 g_m + 2}$
K-HP: $\frac{R_4 R_5 g_m - R_4}{2R_4 g_m + 2R_5 g_m + 2}$
K-BP: 0
Qz: None
Wz: $\frac{1}{\sqrt{C_4} \sqrt{L_4}}$

5 GE

5.1 GE-1 $Z(s) = \left(\infty, \infty, \infty, R_4, L_5 s + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{C_5 L_5 R_4 g_m s^2 - C_5 R_4 s + R_4 g_m}{2C_5 L_5 g_m s^2 + 2g_m + s (2C_5 R_4 g_m + 2C_5)}$$

Parameters:

Q: $\frac{\sqrt{L_5} g_m}{\sqrt{C_5} R_4 g_m + \sqrt{C_5}}$
wo: $\frac{1}{\sqrt{C_5} \sqrt{L_5}}$
bandwidth: $\frac{\sqrt{C_5} R_4 g_m + \sqrt{C_5}}{\sqrt{C_5} L_5 g_m}$
K-LP: $\frac{R_4}{2}$
K-HP: $\frac{R_4}{2}$
K-BP: $-\frac{R_4}{2R_4 g_m + 2}$
Qz: $-\frac{\sqrt{L_5} g_m}{\sqrt{C_5}}$
Wz: $\frac{1}{\sqrt{C_5} \sqrt{L_5}}$

5.2 GE-2 $Z(s) = \left(\infty, \infty, \infty, R_4, \frac{L_5 s}{C_5 L_5 s^2 + 1}, \infty \right)$

$$H(s) = \frac{-C_5 L_5 R_4 s^2 + L_5 R_4 g_m s - R_4}{2L_5 g_m s + 2R_4 g_m + s^2 (2C_5 L_5 R_4 g_m + 2C_5 L_5) + 2}$$

Parameters:

Q: $\frac{\sqrt{C_5} R_4 g_m + \sqrt{C_5}}{\sqrt{L_5} g_m}$
wo: $\frac{1}{\sqrt{C_5} \sqrt{L_5}}$
bandwidth: $\frac{g_m}{\sqrt{C_5} (\sqrt{C_5} R_4 g_m + \sqrt{C_5})}$
K-LP: $-\frac{R_4}{2R_4 g_m + 2}$
K-HP: $-\frac{R_4}{2R_4 g_m + 2}$
K-BP: $\frac{R_4}{2}$
Qz: $-\frac{\sqrt{C_5}}{\sqrt{L_5} g_m}$
Wz: $\frac{1}{\sqrt{C_5} \sqrt{L_5}}$

5.3 GE-3 $Z(s) = \left(\infty, \infty, \infty, R_4, L_5 s + R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{C_5 L_5 R_4 g_m s^2 + R_4 g_m + s (C_5 R_4 R_5 g_m - C_5 R_4)}{2 C_5 L_5 g_m s^2 + 2 g_m + s (2 C_5 R_4 g_m + 2 C_5 R_5 g_m + 2 C_5)}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{L_5 g_m}}{\sqrt{C_5 R_4 g_m + \sqrt{C_5 R_5 g_m + \sqrt{C_5}}}} \\ \text{wo: } & \frac{1}{\sqrt{C_5} \sqrt{L_5}} \\ \text{bandwidth: } & \frac{\sqrt{C_5 R_4 g_m + \sqrt{C_5 R_5 g_m + \sqrt{C_5}}}}{\sqrt{C_5 L_5 g_m}} \\ \text{K-LP: } & \frac{R_4}{2} \\ \text{K-HP: } & \frac{R_4}{2} \\ \text{K-BP: } & \frac{R_4 R_5 g_m - R_4}{2 R_4 g_m + 2 R_5 g_m + 2} \\ \text{QZ: } & \frac{\sqrt{L_5 g_m}}{\sqrt{C_5 R_5 g_m - \sqrt{C_5}}} \\ \text{WZ: } & \frac{1}{\sqrt{C_5} \sqrt{L_5}} \end{aligned}$$

5.4 GE-4 $Z(s) = \left(\infty, \infty, \infty, R_4, \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \infty \right)$

$$H(s) = \frac{-C_5 L_5 R_4 R_5 s^2 - R_4 R_5 + s (L_5 R_4 R_5 g_m - L_5 R_4)}{2 R_4 R_5 g_m + 2 R_5 + s^2 (2 C_5 L_5 R_4 R_5 g_m + 2 C_5 L_5 R_5) + s (2 L_5 R_4 g_m + 2 L_5 R_5 g_m + 2 L_5)}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{C_5 R_4 R_5 g_m + \sqrt{C_5 R_5}}}{\sqrt{L_5 R_4 g_m + \sqrt{L_5 R_5 g_m + \sqrt{L_5}}}} \\ \text{wo: } & \frac{1}{\sqrt{C_5} \sqrt{L_5}} \\ \text{bandwidth: } & \frac{\sqrt{L_5 R_4 g_m + \sqrt{L_5 R_5 g_m + \sqrt{L_5}}}}{\sqrt{C_5} \sqrt{L_5} (\sqrt{C_5 R_4 R_5 g_m + \sqrt{C_5 R_5}})} \\ \text{K-LP: } & -\frac{R_4}{2 R_4 g_m + 2} \\ \text{K-HP: } & -\frac{R_4}{2 R_4 g_m + 2} \\ \text{K-BP: } & \frac{R_4 R_5 g_m - R_4}{2 R_4 g_m + 2 R_5 g_m + 2} \\ \text{QZ: } & -\frac{\sqrt{C_5 R_5}}{\sqrt{L_5 R_5 g_m - \sqrt{L_5}}} \\ \text{WZ: } & \frac{1}{\sqrt{C_5} \sqrt{L_5}} \end{aligned}$$

5.5 GE-5 $Z(s) = \left(\infty, \infty, \infty, R_4, \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \infty \right)$

$$H(s) = \frac{L_5 R_4 g_m s + R_4 R_5 g_m - R_4 + s^2 (C_5 L_5 R_4 R_5 g_m - C_5 L_5 R_4)}{2 L_5 g_m s + 2 R_4 g_m + 2 R_5 g_m + s^2 (2 C_5 L_5 R_4 g_m + 2 C_5 L_5 R_5 g_m + 2 C_5 L_5) + 2}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{C_5 R_4 g_m + \sqrt{C_5 R_5 g_m + \sqrt{C_5}}}}{\sqrt{L_5 g_m}} \\ \text{wo: } & \frac{1}{\sqrt{C_5} \sqrt{L_5}} \\ \text{bandwidth: } & \frac{g_m}{\sqrt{C_5} (\sqrt{C_5 R_4 g_m + \sqrt{C_5 R_5 g_m + \sqrt{C_5}}})} \\ \text{K-LP: } & \frac{R_4 R_5 g_m - R_4}{2 R_4 g_m + 2 R_5 g_m + 2} \\ \text{K-HP: } & \frac{R_4 R_5 g_m - R_4}{2 R_4 g_m + 2 R_5 g_m + 2} \\ \text{K-BP: } & \frac{R_4}{2} \\ \text{QZ: } & \frac{\sqrt{C_5 R_5 g_m - \sqrt{C_5}}}{\sqrt{L_5 g_m}} \\ \text{WZ: } & \frac{1}{\sqrt{C_5} \sqrt{L_5}} \end{aligned}$$

5.6 GE-6 $Z(s) = \left(\infty, \infty, \infty, R_4, \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_5 R_4 R_5 s + R_4 R_5 g_m - R_4 + s^2 (C_5 L_5 R_4 R_5 g_m - C_5 L_5 R_4)}{2 R_4 g_m + 2 R_5 g_m + s^2 (2 C_5 L_5 R_4 g_m + 2 C_5 L_5 R_5 g_m + 2 C_5 L_5) + s (2 C_5 R_4 R_5 g_m + 2 C_5 R_5) + 2}$$

Parameters:

$$\begin{aligned}
\text{Q: } & \frac{\sqrt{L_5}R_4g_m + \sqrt{L_5}R_5g_m + \sqrt{L_5}}{\sqrt{C_5}R_4R_5g_m + \sqrt{C_5}R_5} \\
\text{wo: } & \frac{1}{\sqrt{C_5}\sqrt{L_5}} \\
\text{bandwidth: } & \frac{\sqrt{C_5}R_4R_5g_m + \sqrt{C_5}R_5}{\sqrt{C_5}\sqrt{L_5}(\sqrt{L_5}R_4g_m + \sqrt{L_5}R_5g_m + \sqrt{L_5})} \\
\text{K-LP: } & \frac{R_4R_5g_m - R_4}{2R_4g_m + 2R_5g_m + 2} \\
\text{K-HP: } & \frac{R_4R_5g_m - R_4}{2R_4g_m + 2R_5g_m + 2} \\
\text{K-BP: } & -\frac{R_4}{2R_4g_m + 2} \\
\text{QZ: } & \frac{-\sqrt{L_5}R_5g_m + \sqrt{L_5}}{\sqrt{C_5}R_5} \\
\text{WZ: } & \frac{1}{\sqrt{C_5}\sqrt{L_5}}
\end{aligned}$$

5.7 GE-7 $Z(s) = \left(\infty, \infty, \infty, L_4s + R_4 + \frac{1}{C_4s}, R_5, \infty \right)$

$$H(s) = \frac{R_5g_m + s^2(C_4L_4R_5g_m - C_4L_4) + s(C_4R_4R_5g_m - C_4R_4) - 1}{2C_4L_4g_ms^2 + 2g_m + s(2C_4R_4g_m + 2C_4R_5g_m + 2C_4)}$$

Parameters:

$$\begin{aligned}
\text{Q: } & \frac{\sqrt{L_4}g_m}{\sqrt{C_4}R_4g_m + \sqrt{C_4}R_5g_m + \sqrt{C_4}} \\
\text{wo: } & \frac{1}{\sqrt{C_4}\sqrt{L_4}} \\
\text{bandwidth: } & \frac{\sqrt{C_4}R_4g_m + \sqrt{C_4}R_5g_m + \sqrt{C_4}}{\sqrt{C_4}L_4g_m} \\
\text{K-LP: } & \frac{R_5g_m - 1}{2g_m} \\
\text{K-HP: } & \frac{R_5g_m - 1}{2g_m} \\
\text{K-BP: } & \frac{R_4R_5g_m - R_4}{2R_4g_m + 2R_5g_m + 2} \\
\text{QZ: } & \frac{\sqrt{L_4}}{\sqrt{C_4}R_4} \\
\text{WZ: } & \frac{1}{\sqrt{C_4}\sqrt{L_4}}
\end{aligned}$$

5.8 GE-8 $Z(s) = \left(\infty, \infty, \infty, \frac{C_4L_4R_4s^2 + L_4s + R_4}{C_4L_4s^2 + 1}, R_5, \infty \right)$

$$H(s) = \frac{R_4R_5g_m - R_4 + s^2(C_4L_4R_4R_5g_m - C_4L_4R_4) + s(L_4R_5g_m - L_4)}{2L_4g_ms + 2R_4g_m + 2R_5g_m + s^2(2C_4L_4R_4g_m + 2C_4L_4R_5g_m + 2C_4L_4) + 2}$$

Parameters:

$$\begin{aligned}
\text{Q: } & \frac{\sqrt{C_4}R_4g_m + \sqrt{C_4}R_5g_m + \sqrt{C_4}}{\sqrt{L_4}g_m} \\
\text{wo: } & \frac{1}{\sqrt{C_4}\sqrt{L_4}} \\
\text{bandwidth: } & \frac{g_m}{\sqrt{C_4}(\sqrt{C_4}R_4g_m + \sqrt{C_4}R_5g_m + \sqrt{C_4})} \\
\text{K-LP: } & \frac{R_4R_5g_m - R_4}{2R_4g_m + 2R_5g_m + 2} \\
\text{K-HP: } & \frac{R_4R_5g_m - R_4}{2R_4g_m + 2R_5g_m + 2} \\
\text{K-BP: } & \frac{R_5g_m - 1}{2g_m} \\
\text{QZ: } & \frac{\sqrt{C_4}R_4}{\sqrt{L_4}} \\
\text{WZ: } & \frac{1}{\sqrt{C_4}\sqrt{L_4}}
\end{aligned}$$

6 HP

7 LP

8 X-INVALID-NUMER

8.1 X-INVALID-NUMER-1 $Z(s) = \left(\infty, \infty, \infty, \frac{1}{C_4 s}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_5 R_5 s + R_5 g_m - 1}{2C_4 C_5 R_5 s^2 + 2g_m + s(2C_4 R_5 g_m + 2C_4 + 2C_5 R_5 g_m)}$$

Parameters:

Q: $\frac{\sqrt{C_4} \sqrt{C_5} \sqrt{R_5} \sqrt{g_m}}{C_4 R_5 g_m + C_4 + C_5 R_5 g_m}$
 wo: $\frac{\sqrt{g_m}}{\sqrt{C_4} \sqrt{C_5} \sqrt{R_5}}$
 bandwidth: $\frac{C_4 R_5 g_m + C_4 + C_5 R_5 g_m}{C_4 C_5 R_5}$
 K-LP: $\frac{R_5 g_m - 1}{2g_m}$
 K-HP: 0
 K-BP: $-\frac{C_5 R_5}{2C_4 R_5 g_m + 2C_4 + 2C_5 R_5 g_m}$
 Qz: None
 Wz: None

8.2 X-INVALID-NUMER-2 $Z(s) = \left(\infty, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_5 R_4 s + R_4 g_m}{2C_4 C_5 R_4 s^2 + 2g_m + s(2C_4 R_4 g_m + 2C_5 R_4 g_m + 2C_5)}$$

Parameters:

Q: $\frac{\sqrt{C_4} \sqrt{C_5} \sqrt{R_4} \sqrt{g_m}}{C_4 R_4 g_m + C_5 R_4 g_m + C_5}$
 wo: $\frac{\sqrt{g_m}}{\sqrt{C_4} \sqrt{C_5} \sqrt{R_4}}$
 bandwidth: $\frac{C_4 R_4 g_m + C_5 R_4 g_m + C_5}{C_4 C_5 R_4}$
 K-LP: $\frac{R_4}{2}$
 K-HP: 0
 K-BP: $-\frac{C_5 R_4}{2C_4 R_4 g_m + 2C_5 R_4 g_m + 2C_5}$
 Qz: None
 Wz: None

8.3 X-INVALID-NUMER-3 $Z(s) = \left(\infty, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_5 R_4 R_5 s + R_4 R_5 g_m - R_4}{2C_4 C_5 R_4 R_5 s^2 + 2R_4 g_m + 2R_5 g_m + s(2C_4 R_4 R_5 g_m + 2C_4 R_4 + 2C_5 R_4 R_5 g_m + 2C_5 R_5) + 2}$$

Parameters:

Q: $\frac{\sqrt{C_4} \sqrt{C_5} \sqrt{R_4} \sqrt{R_5} \sqrt{R_4 g_m + R_5 g_m + 1}}{C_4 R_4 R_5 g_m + C_4 R_4 + C_5 R_4 R_5 g_m + C_5 R_5}$
 wo: $\frac{\sqrt{R_4 g_m + R_5 g_m + 1}}{\sqrt{C_4} \sqrt{C_5} \sqrt{R_4} \sqrt{R_5}}$
 bandwidth: $\frac{C_4 R_4 R_5 g_m + C_4 R_4 + C_5 R_4 R_5 g_m + C_5 R_5}{C_4 C_5 R_4 R_5}$
 K-LP: $\frac{R_4 R_5 g_m - R_4}{2R_4 g_m + 2R_5 g_m + 2}$
 K-HP: 0
 K-BP: $-\frac{C_5 R_4 R_5}{2C_4 R_4 R_5 g_m + 2C_4 R_4 + 2C_5 R_4 R_5 g_m + 2C_5 R_5}$
 Qz: None
 Wz: None

8.4 X-INVALID-NUMER-4 $Z(s) = \left(\infty, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{R_4 g_m + s(C_5 R_4 R_5 g_m - C_5 R_4)}{2g_m + s^2(2C_4 C_5 R_4 R_5 g_m + 2C_4 C_5 R_4) + s(2C_4 R_4 g_m + 2C_5 R_4 g_m + 2C_5 R_5 g_m + 2C_5)}$$

Parameters:

Q: $\frac{\sqrt{C_4} \sqrt{C_5} \sqrt{R_4} R_5 g_m \sqrt{\frac{g_m}{R_5 g_m + 1}} + \sqrt{C_4} \sqrt{C_5} \sqrt{R_4} \sqrt{\frac{g_m}{R_5 g_m + 1}}}{C_4 R_4 g_m + C_5 R_4 g_m + C_5 R_5 g_m + C_5}$
 wo: $\sqrt{\frac{g_m}{C_4 C_5 R_4 R_5 g_m + C_4 C_5 R_4}}$

bandwidth: $\frac{\sqrt{\frac{g_m}{C_4 C_5 R_4 R_5 g_m + C_4 C_5 R_4}} (C_4 R_4 g_m + C_5 R_4 g_m + C_5 R_5 g_m + C_5)}{\sqrt{C_4} \sqrt{C_5} \sqrt{R_4 R_5 g_m} \sqrt{\frac{g_m}{R_5 g_m + 1}} + \sqrt{C_4} \sqrt{C_5} \sqrt{R_4} \sqrt{\frac{g_m}{R_5 g_m + 1}}}$
K-LP: $\frac{R_4}{2}$
K-HP: 0
K-BP: $\frac{C_5 R_4 R_5 g_m - C_5 R_4}{2 C_4 R_4 g_m + 2 C_5 R_4 g_m + 2 C_5 R_5 g_m + 2 C_5}$
Qz: None
Wz: None

9 X-INVALID-ORDER

9.1 X-INVALID-ORDER-1 $Z(s) = (\infty, \infty, \infty, R_4, R_5, \infty)$

$$H(s) = \frac{R_4 R_5 g_m - R_4}{2 R_4 g_m + 2 R_5 g_m + 2}$$

9.2 X-INVALID-ORDER-2 $Z(s) = \left(\infty, \infty, \infty, R_4, \frac{1}{C_5 s}, \infty\right)$

$$H(s) = \frac{-C_5 R_4 s + R_4 g_m}{2 g_m + s (2 C_5 R_4 g_m + 2 C_5)}$$

9.3 X-INVALID-ORDER-3 $Z(s) = \left(\infty, \infty, \infty, R_4, \frac{R_5}{C_5 R_5 s + 1}, \infty\right)$

$$H(s) = \frac{-C_5 R_4 R_5 s + R_4 R_5 g_m - R_4}{2 R_4 g_m + 2 R_5 g_m + s (2 C_5 R_4 R_5 g_m + 2 C_5 R_5) + 2}$$

9.4 X-INVALID-ORDER-4 $Z(s) = \left(\infty, \infty, \infty, R_4, R_5 + \frac{1}{C_5 s}, \infty\right)$

$$H(s) = \frac{R_4 g_m + s (C_5 R_4 R_5 g_m - C_5 R_4)}{2 g_m + s (2 C_5 R_4 g_m + 2 C_5 R_5 g_m + 2 C_5)}$$

9.5 X-INVALID-ORDER-5 $Z(s) = \left(\infty, \infty, \infty, \frac{1}{C_4 s}, R_5, \infty\right)$

$$H(s) = \frac{R_5 g_m - 1}{2 g_m + s (2 C_4 R_5 g_m + 2 C_4)}$$

9.6 X-INVALID-ORDER-6 $Z(s) = \left(\infty, \infty, \infty, \frac{1}{C_4 s}, \frac{1}{C_5 s}, \infty\right)$

$$H(s) = \frac{-C_5 s + g_m}{2 C_4 C_5 s^2 + s (2 C_4 g_m + 2 C_5 g_m)}$$

9.7 X-INVALID-ORDER-7 $Z(s) = \left(\infty, \infty, \infty, \frac{1}{C_4 s}, R_5 + \frac{1}{C_5 s}, \infty\right)$

$$H(s) = \frac{g_m + s (C_5 R_5 g_m - C_5)}{s^2 (2 C_4 C_5 R_5 g_m + 2 C_4 C_5) + s (2 C_4 g_m + 2 C_5 g_m)}$$

9.8 X-INVALID-ORDER-8 $Z(s) = \left(\infty, \infty, \infty, \frac{1}{C_4 s}, L_5 s + \frac{1}{C_5 s}, \infty\right)$

$$H(s) = \frac{C_5 L_5 g_m s^2 - C_5 s + g_m}{2 C_4 C_5 L_5 g_m s^3 + 2 C_4 C_5 s^2 + s (2 C_4 g_m + 2 C_5 g_m)}$$

$$\mathbf{9.9 \quad X-INVALID-ORDER-9} \quad Z(s) = \left(\infty, \infty, \infty, \frac{1}{C_4 s}, \frac{L_5 s}{C_5 L_5 s^2 + 1}, \infty \right)$$

$$H(s) = \frac{-C_5 L_5 s^2 + L_5 g_m s - 1}{2C_4 C_5 L_5 s^3 + 2C_4 s + 2g_m + s^2 (2C_4 L_5 g_m + 2C_5 L_5 g_m)}$$

$$\mathbf{9.10 \quad X-INVALID-ORDER-10} \quad Z(s) = \left(\infty, \infty, \infty, \frac{1}{C_4 s}, L_5 s + R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{C_5 L_5 g_m s^2 + g_m + s (C_5 R_5 g_m - C_5)}{2C_4 C_5 L_5 g_m s^3 + s^2 (2C_4 C_5 R_5 g_m + 2C_4 C_5) + s (2C_4 g_m + 2C_5 g_m)}$$

$$\mathbf{9.11 \quad X-INVALID-ORDER-11} \quad Z(s) = \left(\infty, \infty, \infty, \frac{1}{C_4 s}, \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \infty \right)$$

$$H(s) = \frac{-C_5 L_5 R_5 s^2 - R_5 + s (L_5 R_5 g_m - L_5)}{2C_4 C_5 L_5 R_5 s^3 + 2R_5 g_m + s^2 (2C_4 L_5 R_5 g_m + 2C_4 L_5 + 2C_5 L_5 R_5 g_m) + s (2C_4 R_5 + 2L_5 g_m)}$$

$$\mathbf{9.12 \quad X-INVALID-ORDER-12} \quad Z(s) = \left(\infty, \infty, \infty, \frac{1}{C_4 s}, \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \infty \right)$$

$$H(s) = \frac{L_5 g_m s + R_5 g_m + s^2 (C_5 L_5 R_5 g_m - C_5 L_5) - 1}{2g_m + s^3 (2C_4 C_5 L_5 R_5 g_m + 2C_4 C_5 L_5) + s^2 (2C_4 L_5 g_m + 2C_5 L_5 g_m) + s (2C_4 R_5 g_m + 2C_4)}$$

$$\mathbf{9.13 \quad X-INVALID-ORDER-13} \quad Z(s) = \left(\infty, \infty, \infty, \frac{1}{C_4 s}, \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \infty \right)$$

$$H(s) = \frac{-C_5 R_5 s + R_5 g_m + s^2 (C_5 L_5 R_5 g_m - C_5 L_5) - 1}{2g_m + s^3 (2C_4 C_5 L_5 R_5 g_m + 2C_4 C_5 L_5) + s^2 (2C_4 C_5 R_5 + 2C_5 L_5 g_m) + s (2C_4 R_5 g_m + 2C_4 + 2C_5 R_5 g_m)}$$

$$\mathbf{9.14 \quad X-INVALID-ORDER-14} \quad Z(s) = \left(\infty, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, R_5, \infty \right)$$

$$H(s) = \frac{R_4 R_5 g_m - R_4}{2R_4 g_m + 2R_5 g_m + s (2C_4 R_4 R_5 g_m + 2C_4 R_4) + 2}$$

$$\mathbf{9.15 \quad X-INVALID-ORDER-15} \quad Z(s) = \left(\infty, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, L_5 s + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{C_5 L_5 R_4 g_m s^2 - C_5 R_4 s + R_4 g_m}{2C_4 C_5 L_5 R_4 g_m s^3 + 2g_m + s^2 (2C_4 C_5 R_4 + 2C_5 L_5 g_m) + s (2C_4 R_4 g_m + 2C_5 R_4 g_m + 2C_5)}$$

$$\mathbf{9.16 \quad X-INVALID-ORDER-16} \quad Z(s) = \left(\infty, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, \frac{L_5 s}{C_5 L_5 s^2 + 1}, \infty \right)$$

$$H(s) = \frac{-C_5 L_5 R_4 s^2 + L_5 R_4 g_m s - R_4}{2C_4 C_5 L_5 R_4 s^3 + 2R_4 g_m + s^2 (2C_4 L_5 R_4 g_m + 2C_5 L_5 R_4 g_m + 2C_5 L_5) + s (2C_4 R_4 + 2L_5 g_m) + 2}$$

$$\mathbf{9.17 \quad X-INVALID-ORDER-17} \quad Z(s) = \left(\infty, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, L_5 s + R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{C_5 L_5 R_4 g_m s^2 + R_4 g_m + s (C_5 R_4 R_5 g_m - C_5 R_4)}{2C_4 C_5 L_5 R_4 g_m s^3 + 2g_m + s^2 (2C_4 C_5 R_4 R_5 g_m + 2C_4 C_5 R_4 + 2C_5 L_5 g_m) + s (2C_4 R_4 g_m + 2C_5 R_4 g_m + 2C_5 R_5 g_m + 2C_5)}$$

$$\mathbf{9.18 \quad X-INVALID-ORDER-18} \quad Z(s) = \left(\infty, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \infty \right)$$

$$H(s) = \frac{-C_5 L_5 R_4 R_5 s^2 - R_4 R_5 + s (L_5 R_4 R_5 g_m - L_5 R_4)}{2C_4 C_5 L_5 R_4 R_5 s^3 + 2R_4 R_5 g_m + 2R_5 + s^2 (2C_4 L_5 R_4 R_5 g_m + 2C_4 L_5 R_4 + 2C_5 L_5 R_4 R_5 g_m + 2C_5 L_5 R_5) + s (2C_4 R_4 R_5 + 2L_5 R_4 g_m + 2L_5 R_5 g_m + 2L_5)}$$

$$\mathbf{9.19 \quad X-INVALID-ORDER-19} \quad Z(s) = \left(\infty, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \infty \right)$$

$$H(s) = \frac{L_5 R_4 g_m s + R_4 R_5 g_m - R_4 + s^2 (C_5 L_5 R_4 R_5 g_m - C_5 L_5 R_4)}{2 R_4 g_m + 2 R_5 g_m + s^3 (2 C_4 C_5 L_5 R_4 R_5 g_m + 2 C_4 C_5 L_5 R_4) + s^2 (2 C_4 L_5 R_4 g_m + 2 C_5 L_5 R_4 g_m + 2 C_5 L_5 R_5 g_m + 2 C_5 L_5) + s (2 C_4 R_4 R_5 g_m + 2 C_4 R_4 + 2 L_5 g_m) + 2}$$

$$\mathbf{9.20 \quad X-INVALID-ORDER-20} \quad Z(s) = \left(\infty, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \infty \right)$$

$$H(s) = \frac{-C_5 R_4 R_5 s + R_4 R_5 g_m - R_4 + s^2 (C_5 L_5 R_4 R_5 g_m - C_5 L_5 R_4)}{2 R_4 g_m + 2 R_5 g_m + s^3 (2 C_4 C_5 L_5 R_4 R_5 g_m + 2 C_4 C_5 L_5 R_4) + s^2 (2 C_4 C_5 R_4 R_5 + 2 C_5 L_5 R_4 g_m + 2 C_5 L_5 R_5 g_m + 2 C_5 L_5) + s (2 C_4 R_4 R_5 g_m + 2 C_4 R_4 + 2 C_5 R_4 R_5 g_m + 2 C_5 R_5) + 2}$$

$$\mathbf{9.21 \quad X-INVALID-ORDER-21} \quad Z(s) = \left(\infty, \infty, \infty, R_4 + \frac{1}{C_4 s}, R_5, \infty \right)$$

$$H(s) = \frac{R_5 g_m + s (C_4 R_4 R_5 g_m - C_4 R_4) - 1}{2 g_m + s (2 C_4 R_4 g_m + 2 C_4 R_5 g_m + 2 C_4)}$$

$$\mathbf{9.22 \quad X-INVALID-ORDER-22} \quad Z(s) = \left(\infty, \infty, \infty, R_4 + \frac{1}{C_4 s}, \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{-C_4 C_5 R_4 s^2 + g_m + s (C_4 R_4 g_m - C_5)}{s^2 (2 C_4 C_5 R_4 g_m + 2 C_4 C_5) + s (2 C_4 g_m + 2 C_5 g_m)}$$

$$\mathbf{9.23 \quad X-INVALID-ORDER-23} \quad Z(s) = \left(\infty, \infty, \infty, R_4 + \frac{1}{C_4 s}, R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{g_m + s^2 (C_4 C_5 R_4 R_5 g_m - C_4 C_5 R_4) + s (C_4 R_4 g_m + C_5 R_5 g_m - C_5)}{s^2 (2 C_4 C_5 R_4 g_m + 2 C_4 C_5 R_5 g_m + 2 C_4 C_5) + s (2 C_4 g_m + 2 C_5 g_m)}$$

$$\mathbf{9.24 \quad X-INVALID-ORDER-24} \quad Z(s) = \left(\infty, \infty, \infty, R_4 + \frac{1}{C_4 s}, L_5 s + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{C_4 C_5 L_5 R_4 g_m s^3 + g_m + s^2 (-C_4 C_5 R_4 + C_5 L_5 g_m) + s (C_4 R_4 g_m - C_5)}{2 C_4 C_5 L_5 g_m s^3 + s^2 (2 C_4 C_5 R_4 g_m + 2 C_4 C_5) + s (2 C_4 g_m + 2 C_5 g_m)}$$

$$\mathbf{9.25 \quad X-INVALID-ORDER-25} \quad Z(s) = \left(\infty, \infty, \infty, R_4 + \frac{1}{C_4 s}, \frac{L_5 s}{C_5 L_5 s^2 + 1}, \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_5 R_4 s^3 + s^2 (C_4 L_5 R_4 g_m - C_5 L_5) + s (-C_4 R_4 + L_5 g_m) - 1}{2 g_m + s^3 (2 C_4 C_5 L_5 R_4 g_m + 2 C_4 C_5 L_5) + s^2 (2 C_4 L_5 g_m + 2 C_5 L_5 g_m) + s (2 C_4 R_4 g_m + 2 C_4)}$$

$$\mathbf{9.26 \quad X-INVALID-ORDER-26} \quad Z(s) = \left(\infty, \infty, \infty, R_4 + \frac{1}{C_4 s}, L_5 s + R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{C_4 C_5 L_5 R_4 g_m s^3 + g_m + s^2 (C_4 C_5 R_4 R_5 g_m - C_4 C_5 R_4 + C_5 L_5 g_m) + s (C_4 R_4 g_m + C_5 R_5 g_m - C_5)}{2 C_4 C_5 L_5 g_m s^3 + s^2 (2 C_4 C_5 R_4 g_m + 2 C_4 C_5 R_5 g_m + 2 C_4 C_5) + s (2 C_4 g_m + 2 C_5 g_m)}$$

$$\mathbf{9.27 \quad X-INVALID-ORDER-27} \quad Z(s) = \left(\infty, \infty, \infty, R_4 + \frac{1}{C_4 s}, \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_5 R_4 R_5 s^3 - R_5 + s^2 (C_4 L_5 R_4 R_5 g_m - C_4 L_5 R_4 - C_5 L_5 R_5) + s (-C_4 R_4 R_5 + L_5 R_5 g_m - L_5)}{2 R_5 g_m + s^3 (2 C_4 C_5 L_5 R_4 R_5 g_m + 2 C_4 C_5 L_5 R_5) + s^2 (2 C_4 L_5 R_4 g_m + 2 C_4 L_5 R_5 g_m + 2 C_4 L_5 + 2 C_5 L_5 R_5 g_m) + s (2 C_4 R_4 R_5 g_m + 2 C_4 R_5 + 2 L_5 g_m)}$$

$$\mathbf{9.28 \quad X-INVALID-ORDER-28} \quad Z(s) = \left(\infty, \infty, \infty, R_4 + \frac{1}{C_4 s}, \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \infty \right)$$

$$H(s) = \frac{R_5 g_m + s^3 (C_4 C_5 L_5 R_4 R_5 g_m - C_4 C_5 L_5 R_4) + s^2 (C_4 L_5 R_4 g_m + C_5 L_5 R_5 g_m - C_5 L_5) + s (C_4 R_4 R_5 g_m - C_4 R_4 + L_5 g_m) - 1}{2g_m + s^3 (2C_4 C_5 L_5 R_4 g_m + 2C_4 C_5 L_5 R_5 g_m + 2C_4 C_5 L_5) + s^2 (2C_4 L_5 g_m + 2C_5 L_5 g_m) + s (2C_4 R_4 g_m + 2C_4 R_5 g_m + 2C_4)}$$

$$\mathbf{9.29 \quad X-INVALID-ORDER-29} \quad Z(s) = \left(\infty, \infty, \infty, R_4 + \frac{1}{C_4 s}, \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \infty \right)$$

$$H(s) = \frac{R_5 g_m + s^3 (C_4 C_5 L_5 R_4 R_5 g_m - C_4 C_5 L_5 R_4) + s^2 (-C_4 C_5 R_4 R_5 + C_5 L_5 R_5 g_m - C_5 L_5) + s (C_4 R_4 R_5 g_m - C_4 R_4 - C_5 R_5) - 1}{2g_m + s^3 (2C_4 C_5 L_5 R_4 g_m + 2C_4 C_5 L_5 R_5 g_m + 2C_4 C_5 L_5) + s^2 (2C_4 C_5 R_4 R_5 g_m + 2C_4 C_5 R_5 + 2C_5 L_5 g_m) + s (2C_4 R_4 g_m + 2C_4 R_5 g_m + 2C_4 + 2C_5 R_5 g_m)}$$

$$\mathbf{9.30 \quad X-INVALID-ORDER-30} \quad Z(s) = \left(\infty, \infty, \infty, L_4 s + \frac{1}{C_4 s}, \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_4 s^3 + C_4 L_4 g_m s^2 - C_5 s + g_m}{2C_4 C_5 L_4 g_m s^3 + 2C_4 C_5 s^2 + s (2C_4 g_m + 2C_5 g_m)}$$

$$\mathbf{9.31 \quad X-INVALID-ORDER-31} \quad Z(s) = \left(\infty, \infty, \infty, L_4 s + \frac{1}{C_4 s}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_4 R_5 s^3 - C_5 R_5 s + R_5 g_m + s^2 (C_4 L_4 R_5 g_m - C_4 L_4) - 1}{2C_4 C_5 L_4 R_5 g_m s^3 + 2g_m + s^2 (2C_4 C_5 R_5 + 2C_4 L_4 g_m) + s (2C_4 R_5 g_m + 2C_4 + 2C_5 R_5 g_m)}$$

$$\mathbf{9.32 \quad X-INVALID-ORDER-32} \quad Z(s) = \left(\infty, \infty, \infty, L_4 s + \frac{1}{C_4 s}, R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{C_4 L_4 g_m s^2 + g_m + s^3 (C_4 C_5 L_4 R_5 g_m - C_4 C_5 L_4) + s (C_5 R_5 g_m - C_5)}{2C_4 C_5 L_4 g_m s^3 + s^2 (2C_4 C_5 R_5 g_m + 2C_4 C_5) + s (2C_4 g_m + 2C_5 g_m)}$$

$$\mathbf{9.33 \quad X-INVALID-ORDER-33} \quad Z(s) = \left(\infty, \infty, \infty, L_4 s + \frac{1}{C_4 s}, L_5 s + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{C_4 C_5 L_4 L_5 g_m s^4 - C_4 C_5 L_4 s^3 - C_5 s + g_m + s^2 (C_4 L_4 g_m + C_5 L_5 g_m)}{2C_4 C_5 s^2 + s^3 (2C_4 C_5 L_4 g_m + 2C_4 C_5 L_5 g_m) + s (2C_4 g_m + 2C_5 g_m)}$$

$$\mathbf{9.34 \quad X-INVALID-ORDER-34} \quad Z(s) = \left(\infty, \infty, \infty, L_4 s + \frac{1}{C_4 s}, \frac{L_5 s}{C_5 L_5 s^2 + 1}, \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_4 L_5 s^4 + C_4 L_4 L_5 g_m s^3 + L_5 g_m s + s^2 (-C_4 L_4 - C_5 L_5) - 1}{2C_4 C_5 L_4 L_5 g_m s^4 + 2C_4 C_5 L_5 s^3 + 2C_4 s + 2g_m + s^2 (2C_4 L_4 g_m + 2C_4 L_5 g_m + 2C_5 L_5 g_m)}$$

$$\mathbf{9.35 \quad X-INVALID-ORDER-35} \quad Z(s) = \left(\infty, \infty, \infty, L_4 s + \frac{1}{C_4 s}, L_5 s + R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{C_4 C_5 L_4 L_5 g_m s^4 + g_m + s^3 (C_4 C_5 L_4 R_5 g_m - C_4 C_5 L_4) + s^2 (C_4 L_4 g_m + C_5 L_5 g_m) + s (C_5 R_5 g_m - C_5)}{s^3 (2C_4 C_5 L_4 g_m + 2C_4 C_5 L_5 g_m) + s^2 (2C_4 C_5 R_5 g_m + 2C_4 C_5) + s (2C_4 g_m + 2C_5 g_m)}$$

$$\mathbf{9.36 \quad X-INVALID-ORDER-36} \quad Z(s) = \left(\infty, \infty, \infty, L_4 s + \frac{1}{C_4 s}, \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_4 L_5 R_5 s^4 - R_5 + s^3 (C_4 L_4 L_5 R_5 g_m - C_4 L_4 L_5) + s^2 (-C_4 L_4 R_5 - C_5 L_5 R_5) + s (L_5 R_5 g_m - L_5)}{2C_4 C_5 L_4 L_5 R_5 g_m s^4 + 2R_5 g_m + s^3 (2C_4 C_5 L_5 R_5 + 2C_4 L_4 L_5 g_m) + s^2 (2C_4 L_4 R_5 g_m + 2C_4 L_5 R_5 g_m + 2C_4 L_5 + 2C_5 L_5 R_5 g_m) + s (2C_4 R_5 + 2L_5 g_m)}$$

$$\mathbf{9.37 \quad X-INVALID-ORDER-37} \quad Z(s) = \left(\infty, \infty, \infty, L_4 s + \frac{1}{C_4 s}, \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \infty \right)$$

$$H(s) = \frac{C_4 L_4 L_5 g_m s^3 + L_5 g_m s + R_5 g_m + s^4 (C_4 C_5 L_4 L_5 R_5 g_m - C_4 C_5 L_4 L_5) + s^2 (C_4 L_4 R_5 g_m - C_4 L_4 + C_5 L_5 R_5 g_m - C_5 L_5) - 1}{2 C_4 C_5 L_4 L_5 g_m s^4 + 2 g_m + s^3 (2 C_4 C_5 L_5 R_5 g_m + 2 C_4 C_5 L_5) + s^2 (2 C_4 L_4 g_m + 2 C_4 L_5 g_m + 2 C_5 L_5 g_m) + s (2 C_4 R_5 g_m + 2 C_4)}$$

$$\mathbf{9.38 \quad X-INVALID-ORDER-38} \quad Z(s) = \left(\infty, \infty, \infty, L_4 s + \frac{1}{C_4 s}, \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_4 R_5 s^3 - C_5 R_5 s + R_5 g_m + s^4 (C_4 C_5 L_4 L_5 R_5 g_m - C_4 C_5 L_4 L_5) + s^2 (C_4 L_4 R_5 g_m - C_4 L_4 + C_5 L_5 R_5 g_m - C_5 L_5) - 1}{2 C_4 C_5 L_4 L_5 g_m s^4 + 2 g_m + s^3 (2 C_4 C_5 L_4 R_5 g_m + 2 C_4 C_5 L_5 R_5 g_m + 2 C_4 C_5 L_5) + s^2 (2 C_4 C_5 R_5 + 2 C_4 L_4 g_m + 2 C_5 L_5 g_m) + s (2 C_4 R_5 g_m + 2 C_4 + 2 C_5 R_5 g_m)}$$

$$\mathbf{9.39 \quad X-INVALID-ORDER-39} \quad Z(s) = \left(\infty, \infty, \infty, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{-C_5 L_4 s^2 + L_4 g_m s}{2 C_4 C_5 L_4 s^3 + 2 C_5 s + 2 g_m + s^2 (2 C_4 L_4 g_m + 2 C_5 L_4 g_m)}$$

$$\mathbf{9.40 \quad X-INVALID-ORDER-40} \quad Z(s) = \left(\infty, \infty, \infty, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$$

$$H(s) = \frac{-C_5 L_4 R_5 s^2 + s (L_4 R_5 g_m - L_4)}{2 C_4 C_5 L_4 R_5 s^3 + 2 R_5 g_m + s^2 (2 C_4 L_4 R_5 g_m + 2 C_4 L_4 + 2 C_5 L_4 R_5 g_m) + s (2 C_5 R_5 + 2 L_4 g_m) + 2}$$

$$\mathbf{9.41 \quad X-INVALID-ORDER-41} \quad Z(s) = \left(\infty, \infty, \infty, \frac{L_4 s}{C_4 L_4 s^2 + 1}, R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{L_4 g_m s + s^2 (C_5 L_4 R_5 g_m - C_5 L_4)}{2 g_m + s^3 (2 C_4 C_5 L_4 R_5 g_m + 2 C_4 C_5 L_4) + s^2 (2 C_4 L_4 g_m + 2 C_5 L_4 g_m) + s (2 C_5 R_5 g_m + 2 C_5)}$$

$$\mathbf{9.42 \quad X-INVALID-ORDER-42} \quad Z(s) = \left(\infty, \infty, \infty, \frac{L_4 s}{C_4 L_4 s^2 + 1}, L_5 s + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{C_5 L_4 L_5 g_m s^3 - C_5 L_4 s^2 + L_4 g_m s}{2 C_4 C_5 L_4 L_5 g_m s^4 + 2 C_4 C_5 L_4 s^3 + 2 C_5 s + 2 g_m + s^2 (2 C_4 L_4 g_m + 2 C_5 L_4 g_m + 2 C_5 L_5 g_m)}$$

$$\mathbf{9.43 \quad X-INVALID-ORDER-43} \quad Z(s) = \left(\infty, \infty, \infty, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \frac{L_5 s}{C_5 L_5 s^2 + 1}, \infty \right)$$

$$H(s) = \frac{-C_5 L_4 L_5 s^3 + L_4 L_5 g_m s^2 - L_4 s}{2 C_4 C_5 L_4 L_5 s^4 + s^3 (2 C_4 L_4 L_5 g_m + 2 C_5 L_4 L_5 g_m) + s^2 (2 C_4 L_4 + 2 C_5 L_5) + s (2 L_4 g_m + 2 L_5 g_m) + 2}$$

$$\mathbf{9.44 \quad X-INVALID-ORDER-44} \quad Z(s) = \left(\infty, \infty, \infty, \frac{L_4 s}{C_4 L_4 s^2 + 1}, L_5 s + R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{C_5 L_4 L_5 g_m s^3 + L_4 g_m s + s^2 (C_5 L_4 R_5 g_m - C_5 L_4)}{2 C_4 C_5 L_4 L_5 g_m s^4 + 2 g_m + s^3 (2 C_4 C_5 L_4 R_5 g_m + 2 C_4 C_5 L_4) + s^2 (2 C_4 L_4 g_m + 2 C_5 L_4 g_m + 2 C_5 L_5 g_m) + s (2 C_5 R_5 g_m + 2 C_5)}$$

$$\mathbf{9.45 \quad X-INVALID-ORDER-45} \quad Z(s) = \left(\infty, \infty, \infty, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \infty \right)$$

$$H(s) = \frac{-C_5 L_4 L_5 R_5 s^3 - L_4 R_5 s + s^2 (L_4 L_5 R_5 g_m - L_4 L_5)}{2 C_4 C_5 L_4 L_5 R_5 s^4 + 2 R_5 + s^3 (2 C_4 L_4 L_5 R_5 g_m + 2 C_4 L_4 L_5 + 2 C_5 L_4 L_5 R_5 g_m) + s^2 (2 C_4 L_4 R_5 + 2 C_5 L_5 R_5 + 2 L_4 L_5 g_m) + s (2 L_4 R_5 g_m + 2 L_5 R_5 g_m + 2 L_5)}$$

$$9.46 \quad \text{X-INVALID-ORDER-46} \quad Z(s) = \left(\infty, \infty, \infty, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \infty \right)$$

$$H(s) = \frac{L_4 L_5 g_m s^2 + s^3 (C_5 L_4 L_5 R_5 g_m - C_5 L_4 L_5) + s (L_4 R_5 g_m - L_4)}{2 R_5 g_m + s^4 (2 C_4 C_5 L_4 L_5 R_5 g_m + 2 C_4 C_5 L_4 L_5) + s^3 (2 C_4 L_4 L_5 g_m + 2 C_5 L_4 L_5 g_m) + s^2 (2 C_4 L_4 R_5 g_m + 2 C_4 L_4 + 2 C_5 L_5 R_5 g_m + 2 C_5 L_5) + s (2 L_4 g_m + 2 L_5 g_m) + 2}$$

$$9.47 \quad \text{X-INVALID-ORDER-47} \quad Z(s) = \left(\infty, \infty, \infty, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \infty \right)$$

$$H(s) = \frac{-C_5 L_4 R_5 s^2 + s^3 (C_5 L_4 L_5 R_5 g_m - C_5 L_4 L_5) + s (L_4 R_5 g_m - L_4)}{2 R_5 g_m + s^4 (2 C_4 C_5 L_4 L_5 R_5 g_m + 2 C_4 C_5 L_4 L_5) + s^3 (2 C_4 C_5 L_4 R_5 + 2 C_5 L_4 L_5 g_m) + s^2 (2 C_4 L_4 R_5 g_m + 2 C_4 L_4 + 2 C_5 L_4 R_5 g_m + 2 C_5 L_5 R_5 g_m + 2 C_5 L_5) + s (2 C_5 R_5 + 2 L_4 g_m) + 2}$$

$$9.48 \quad \text{X-INVALID-ORDER-48} \quad Z(s) = \left(\infty, \infty, \infty, L_4 s + R_4 + \frac{1}{C_4 s}, \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_4 s^3 + g_m + s^2 (-C_4 C_5 R_4 + C_4 L_4 g_m) + s (C_4 R_4 g_m - C_5)}{2 C_4 C_5 L_4 g_m s^3 + s^2 (2 C_4 C_5 R_4 g_m + 2 C_4 C_5) + s (2 C_4 g_m + 2 C_5 g_m)}$$

$$9.49 \quad \text{X-INVALID-ORDER-49} \quad Z(s) = \left(\infty, \infty, \infty, L_4 s + R_4 + \frac{1}{C_4 s}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_4 R_5 s^3 + R_5 g_m + s^2 (-C_4 C_5 R_4 R_5 + C_4 L_4 R_5 g_m - C_4 L_4) + s (C_4 R_4 R_5 g_m - C_4 R_4 - C_5 R_5) - 1}{2 C_4 C_5 L_4 R_5 g_m s^3 + 2 g_m + s^2 (2 C_4 C_5 R_4 R_5 g_m + 2 C_4 C_5 R_5 + 2 C_4 L_4 g_m) + s (2 C_4 R_4 g_m + 2 C_4 R_5 g_m + 2 C_4 + 2 C_5 R_5 g_m)}$$

$$9.50 \quad \text{X-INVALID-ORDER-50} \quad Z(s) = \left(\infty, \infty, \infty, L_4 s + R_4 + \frac{1}{C_4 s}, R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{g_m + s^3 (C_4 C_5 L_4 R_5 g_m - C_4 C_5 L_4) + s^2 (C_4 C_5 R_4 R_5 g_m - C_4 C_5 R_4 + C_4 L_4 g_m) + s (C_4 R_4 g_m + C_5 R_5 g_m - C_5)}{2 C_4 C_5 L_4 g_m s^3 + s^2 (2 C_4 C_5 R_4 g_m + 2 C_4 C_5 R_5 g_m + 2 C_4 C_5) + s (2 C_4 g_m + 2 C_5 g_m)}$$

$$9.51 \quad \text{X-INVALID-ORDER-51} \quad Z(s) = \left(\infty, \infty, \infty, L_4 s + R_4 + \frac{1}{C_4 s}, L_5 s + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{C_4 C_5 L_4 L_5 g_m s^4 + g_m + s^3 (-C_4 C_5 L_4 + C_4 C_5 L_5 R_4 g_m) + s^2 (-C_4 C_5 R_4 + C_4 L_4 g_m + C_5 L_5 g_m) + s (C_4 R_4 g_m - C_5)}{s^3 (2 C_4 C_5 L_4 g_m + 2 C_4 C_5 L_5 g_m) + s^2 (2 C_4 C_5 R_4 g_m + 2 C_4 C_5) + s (2 C_4 g_m + 2 C_5 g_m)}$$

$$9.52 \quad \text{X-INVALID-ORDER-52} \quad Z(s) = \left(\infty, \infty, \infty, L_4 s + R_4 + \frac{1}{C_4 s}, \frac{L_5 s}{C_5 L_5 s^2 + 1}, \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_4 L_5 s^4 + s^3 (-C_4 C_5 L_5 R_4 + C_4 L_4 L_5 g_m) + s^2 (-C_4 L_4 + C_4 L_5 R_4 g_m - C_5 L_5) + s (-C_4 R_4 + L_5 g_m) - 1}{2 C_4 C_5 L_4 L_5 g_m s^4 + 2 g_m + s^3 (2 C_4 C_5 L_5 R_4 g_m + 2 C_4 C_5 L_5) + s^2 (2 C_4 L_4 g_m + 2 C_4 L_5 g_m + 2 C_5 L_5 g_m) + s (2 C_4 R_4 g_m + 2 C_4)}$$

$$9.53 \quad \text{X-INVALID-ORDER-53} \quad Z(s) = \left(\infty, \infty, \infty, L_4 s + R_4 + \frac{1}{C_4 s}, L_5 s + R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{C_4 C_5 L_4 L_5 g_m s^4 + g_m + s^3 (C_4 C_5 L_4 R_5 g_m - C_4 C_5 L_4 + C_4 C_5 L_5 R_4 g_m) + s^2 (C_4 C_5 R_4 R_5 g_m - C_4 C_5 R_4 + C_4 L_4 g_m + C_5 L_5 g_m) + s (C_4 R_4 g_m + C_5 R_5 g_m - C_5)}{s^3 (2 C_4 C_5 L_4 g_m + 2 C_4 C_5 L_5 g_m) + s^2 (2 C_4 C_5 R_4 g_m + 2 C_4 C_5 R_5 g_m + 2 C_4 C_5) + s (2 C_4 g_m + 2 C_5 g_m)}$$

$$9.54 \quad \text{X-INVALID-ORDER-54} \quad Z(s) = \left(\infty, \infty, \infty, L_4 s + R_4 + \frac{1}{C_4 s}, \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_4 L_5 R_5 s^4 - R_5 + s^3 (-C_4 C_5 L_5 R_4 R_5 + C_4 L_4 L_5 R_5 g_m - C_4 L_4 L_5) + s^2 (-C_4 L_4 R_5 + C_4 L_5 R_4 R_5 g_m - C_4 L_5 R_4 - C_5 L_5 R_5) + s (-C_4 R_4 R_5 + L_5 R_5 g_m - L_5)}{2 C_4 C_5 L_4 L_5 R_5 g_m s^4 + 2 R_5 g_m + s^3 (2 C_4 C_5 L_5 R_4 R_5 g_m + 2 C_4 C_5 L_5 R_5 + 2 C_4 L_4 L_5 g_m) + s^2 (2 C_4 L_4 R_5 g_m + 2 C_4 L_5 R_4 g_m + 2 C_4 L_5 R_5 g_m + 2 C_4 L_5 + 2 C_5 L_5 R_5 g_m) + s (2 C_4 R_4 R_5 g_m + 2 C_4 R_5 + 2 L_5 g_m)}$$

$$\mathbf{9.55 \quad X-INVALID-ORDER-55} \quad Z(s) = \left(\infty, \infty, \infty, L_4 s + R_4 + \frac{1}{C_4 s}, \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \infty \right)$$

$$H(s) = \frac{R_5 g_m + s^4 (C_4 C_5 L_4 L_5 R_5 g_m - C_4 C_5 L_4 L_5) + s^3 (C_4 C_5 L_5 R_4 R_5 g_m - C_4 C_5 L_5 R_4 + C_4 L_4 L_5 g_m) + s^2 (C_4 L_4 R_5 g_m - C_4 L_4 + C_4 L_5 R_4 g_m + C_5 L_5 R_5 g_m - C_5 L_5) + s (C_4 R_4 R_5 g_m - C_4 R_4 + L_5 g_m) - 1}{2 C_4 C_5 L_4 L_5 g_m s^4 + 2 g_m + s^3 (2 C_4 C_5 L_5 R_4 g_m + 2 C_4 C_5 L_5 R_5 g_m + 2 C_4 C_5 L_5) + s^2 (2 C_4 L_4 g_m + 2 C_4 L_5 g_m + 2 C_5 L_5 g_m) + s (2 C_4 R_4 g_m + 2 C_4 R_5 g_m + 2 C_4)}$$

$$\mathbf{9.56 \quad X-INVALID-ORDER-56} \quad Z(s) = \left(\infty, \infty, \infty, L_4 s + R_4 + \frac{1}{C_4 s}, \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \infty \right)$$

$$H(s) = \frac{R_5 g_m + s^4 (C_4 C_5 L_4 L_5 R_5 g_m - C_4 C_5 L_4 L_5) + s^3 (-C_4 C_5 L_4 R_5 + C_4 C_5 L_5 R_4 R_5 g_m - C_4 C_5 L_5 R_4) + s^2 (-C_4 C_5 R_4 R_5 + C_4 L_4 R_5 g_m - C_4 L_4 + C_5 L_5 R_5 g_m - C_5 L_5) + s (C_4 R_4 R_5 g_m - C_4 R_4 - C_5 R_5) - 1}{2 C_4 C_5 L_4 L_5 g_m s^4 + 2 g_m + s^3 (2 C_4 C_5 L_4 R_5 g_m + 2 C_4 C_5 L_5 R_4 g_m + 2 C_4 C_5 L_5 R_5 g_m + 2 C_4 C_5 L_5) + s^2 (2 C_4 C_5 R_4 R_5 g_m + 2 C_4 C_5 R_5 + 2 C_4 L_4 g_m + 2 C_5 L_5 g_m) + s (2 C_4 R_4 g_m + 2 C_4 R_5 g_m + 2 C_4 + 2 C_5 R_5 g_m)}$$

$$\mathbf{9.57 \quad X-INVALID-ORDER-57} \quad Z(s) = \left(\infty, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{-C_5 L_4 R_4 s^2 + L_4 R_4 g_m s}{2 C_4 C_5 L_4 R_4 s^3 + 2 R_4 g_m + s^2 (2 C_4 L_4 R_4 g_m + 2 C_5 L_4 R_4 g_m + 2 C_5 L_4) + s (2 C_5 R_4 + 2 L_4 g_m)}$$

$$\mathbf{9.58 \quad X-INVALID-ORDER-58} \quad Z(s) = \left(\infty, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$$

$$H(s) = \frac{-C_5 L_4 R_4 R_5 s^2 + s (L_4 R_4 R_5 g_m - L_4 R_4)}{2 C_4 C_5 L_4 R_4 R_5 s^3 + 2 R_4 R_5 g_m + 2 R_4 + s^2 (2 C_4 L_4 R_4 R_5 g_m + 2 C_4 L_4 R_4 + 2 C_5 L_4 R_4 R_5 g_m + 2 C_5 L_4 R_5) + s (2 C_5 R_4 R_5 + 2 L_4 R_4 g_m + 2 L_4 R_5 g_m + 2 L_4)}$$

$$\mathbf{9.59 \quad X-INVALID-ORDER-59} \quad Z(s) = \left(\infty, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{L_4 R_4 g_m s + s^2 (C_5 L_4 R_4 R_5 g_m - C_5 L_4 R_4)}{2 R_4 g_m + s^3 (2 C_4 C_5 L_4 R_4 R_5 g_m + 2 C_4 C_5 L_4 R_4) + s^2 (2 C_4 L_4 R_4 g_m + 2 C_5 L_4 R_4 g_m + 2 C_5 L_4 R_5 g_m + 2 C_5 L_4) + s (2 C_5 R_4 R_5 g_m + 2 C_5 R_4 + 2 L_4 g_m)}$$

$$\mathbf{9.60 \quad X-INVALID-ORDER-60} \quad Z(s) = \left(\infty, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, L_5 s + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{C_5 L_4 L_5 R_4 g_m s^3 - C_5 L_4 R_4 s^2 + L_4 R_4 g_m s}{2 C_4 C_5 L_4 L_5 R_4 g_m s^4 + 2 R_4 g_m + s^3 (2 C_4 C_5 L_4 R_4 + 2 C_5 L_4 L_5 g_m) + s^2 (2 C_4 L_4 R_4 g_m + 2 C_5 L_4 R_4 g_m + 2 C_5 L_4 + 2 C_5 L_5 R_4 g_m) + s (2 C_5 R_4 + 2 L_4 g_m)}$$

$$\mathbf{9.61 \quad X-INVALID-ORDER-61} \quad Z(s) = \left(\infty, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \frac{L_5 s}{C_5 L_5 s^2 + 1}, \infty \right)$$

$$H(s) = \frac{-C_5 L_4 L_5 R_4 s^3 + L_4 L_5 R_4 g_m s^2 - L_4 R_4 s}{2 C_4 C_5 L_4 L_5 R_4 s^4 + 2 R_4 + s^3 (2 C_4 L_4 L_5 R_4 g_m + 2 C_5 L_4 L_5 R_4 g_m + 2 C_5 L_4 L_5) + s^2 (2 C_4 L_4 R_4 + 2 C_5 L_5 R_4 + 2 L_4 L_5 g_m) + s (2 L_4 R_4 g_m + 2 L_4 + 2 L_5 R_4 g_m)}$$

$$\mathbf{9.62 \quad X-INVALID-ORDER-62} \quad Z(s) = \left(\infty, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, L_5 s + R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{C_5 L_4 L_5 R_4 g_m s^3 + L_4 R_4 g_m s + s^2 (C_5 L_4 R_4 R_5 g_m - C_5 L_4 R_4)}{2 C_4 C_5 L_4 L_5 R_4 g_m s^4 + 2 R_4 g_m + s^3 (2 C_4 C_5 L_4 R_4 R_5 g_m + 2 C_4 C_5 L_4 R_4 + 2 C_5 L_4 L_5 g_m) + s^2 (2 C_4 L_4 R_4 g_m + 2 C_5 L_4 R_4 g_m + 2 C_5 L_4 R_5 g_m + 2 C_5 L_4 + 2 C_5 L_5 R_4 g_m) + s (2 C_5 R_4 R_5 g_m + 2 C_5 R_4 + 2 L_4 g_m)}$$

$$\mathbf{9.63 \quad X-INVALID-ORDER-63} \quad Z(s) = \left(\infty, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \infty \right)$$

$$H(s) = \frac{-C_5 L_4 L_5 R_4 R_5 s^3 - L_4 R_4 R_5 s + s^2 (L_4 L_5 R_4 R_5 g_m - L_4 L_5 R_4)}{2 C_4 C_5 L_4 L_5 R_4 R_5 s^4 + 2 R_4 R_5 + s^3 (2 C_4 L_4 L_5 R_4 R_5 g_m + 2 C_4 L_4 L_5 R_4 + 2 C_5 L_4 L_5 R_4 R_5 g_m + 2 C_5 L_4 L_5 R_5) + s^2 (2 C_4 L_4 R_4 R_5 + 2 C_5 L_5 R_4 R_5 + 2 L_4 L_5 R_4 g_m + 2 L_4 L_5 R_5 g_m + 2 L_4 L_5) + s (2 L_4 R_4 R_5 g_m + 2 L_4 R_5 + 2 L_5 R_4 R_5 g_m + 2 L_5 R_4)}$$

$$\mathbf{9.64 \quad X-INVALID-ORDER-64} \quad Z(s) = \left(\infty, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \infty \right)$$

$$H(s) = \frac{L_4 L_5 R_4 g_m s^2 + s^3 (C_5 L_4 L_5 R_4 R_5 g_m - C_5 L_4 L_5 R_4) + s (L_4 R_4 R_5 g_m - L_4 R_4)}{2 R_4 R_5 g_m + 2 R_4 + s^4 (2 C_4 C_5 L_4 L_5 R_4 R_5 g_m + 2 C_4 C_5 L_4 L_5 R_4) + s^3 (2 C_4 L_4 L_5 R_4 g_m + 2 C_5 L_4 L_5 R_4 g_m + 2 C_5 L_4 L_5 R_5 g_m + 2 C_5 L_4 L_5) + s^2 (2 C_4 L_4 R_4 R_5 g_m + 2 C_4 L_4 R_4 + 2 C_5 L_5 R_4 R_5 g_m + 2 C_5 L_5 R_4 + 2 L_4 L_5 g_m) + s (2 L_4 R_4 g_m + 2 L_4 R_5 g_m + 2 L_4 + 2 L_5 R_4 g_m)}$$

$$\mathbf{9.65 \quad X-INVALID-ORDER-65} \quad Z(s) = \left(\infty, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \infty \right)$$

$$H(s) = \frac{-C_5 L_4 R_4 R_5 s^2 + s^3 (C_5 L_4 L_5 R_4 R_5 g_m - C_5 L_4 L_5 R_4) + s (L_4 R_4 R_5 g_m - L_4 R_4)}{2 R_4 R_5 g_m + 2 R_4 + s^4 (2 C_4 C_5 L_4 L_5 R_4 R_5 g_m + 2 C_4 C_5 L_4 L_5 R_4) + s^3 (2 C_4 C_5 L_4 R_4 R_5 + 2 C_5 L_4 L_5 R_4 g_m + 2 C_5 L_4 L_5 R_5 g_m + 2 C_5 L_4 L_5) + s^2 (2 C_4 L_4 R_4 R_5 g_m + 2 C_4 L_4 R_4 + 2 C_5 L_4 R_4 R_5 g_m + 2 C_5 L_4 R_5 + 2 C_5 L_5 R_4 R_5 g_m + 2 C_5 L_5 R_4) + s (2 C_5 R_4 R_5 + 2 L_4 R_4 g_m + 2 L_4 R_5)}$$

$$\mathbf{9.66 \quad X-INVALID-ORDER-66} \quad Z(s) = \left(\infty, \infty, \infty, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_4 R_4 s^3 + R_4 g_m + s^2 (C_4 L_4 R_4 g_m - C_5 L_4) + s (-C_5 R_4 + L_4 g_m)}{2 g_m + s^3 (2 C_4 C_5 L_4 R_4 g_m + 2 C_4 C_5 L_4) + s^2 (2 C_4 L_4 g_m + 2 C_5 L_4 g_m) + s (2 C_5 R_4 g_m + 2 C_5)}$$

$$\mathbf{9.67 \quad X-INVALID-ORDER-67} \quad Z(s) = \left(\infty, \infty, \infty, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_4 R_4 R_5 s^3 + R_4 R_5 g_m - R_4 + s^2 (C_4 L_4 R_4 R_5 g_m - C_4 L_4 R_4 - C_5 L_4 R_5) + s (-C_5 R_4 R_5 + L_4 R_5 g_m - L_4)}{2 R_4 g_m + 2 R_5 g_m + s^3 (2 C_4 C_5 L_4 R_4 R_5 g_m + 2 C_4 C_5 L_4 R_5) + s^2 (2 C_4 L_4 R_4 g_m + 2 C_4 L_4 R_5 g_m + 2 C_4 L_4 + 2 C_5 L_4 R_5 g_m) + s (2 C_5 R_4 R_5 g_m + 2 C_5 R_5 + 2 L_4 g_m) + 2}$$

$$\mathbf{9.68 \quad X-INVALID-ORDER-68} \quad Z(s) = \left(\infty, \infty, \infty, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{R_4 g_m + s^3 (C_4 C_5 L_4 R_4 R_5 g_m - C_4 C_5 L_4 R_4) + s^2 (C_4 L_4 R_4 g_m + C_5 L_4 R_5 g_m - C_5 L_4) + s (C_5 R_4 R_5 g_m - C_5 R_4 + L_4 g_m)}{2 g_m + s^3 (2 C_4 C_5 L_4 R_4 g_m + 2 C_4 C_5 L_4 R_5 g_m + 2 C_4 C_5 L_4) + s^2 (2 C_4 L_4 g_m + 2 C_5 L_4 g_m) + s (2 C_5 R_4 g_m + 2 C_5 R_5 g_m + 2 C_5)}$$

$$\mathbf{9.69 \quad X-INVALID-ORDER-69} \quad Z(s) = \left(\infty, \infty, \infty, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, L_5 s + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{C_4 C_5 L_4 L_5 R_4 g_m s^4 + R_4 g_m + s^3 (-C_4 C_5 L_4 R_4 + C_5 L_4 L_5 g_m) + s^2 (C_4 L_4 R_4 g_m - C_5 L_4 + C_5 L_5 R_4 g_m) + s (-C_5 R_4 + L_4 g_m)}{2 C_4 C_5 L_4 L_5 g_m s^4 + 2 g_m + s^3 (2 C_4 C_5 L_4 R_4 g_m + 2 C_4 C_5 L_4) + s^2 (2 C_4 L_4 g_m + 2 C_5 L_4 g_m + 2 C_5 L_5 g_m) + s (2 C_5 R_4 g_m + 2 C_5)}$$

$$\mathbf{9.70 \quad X-INVALID-ORDER-70} \quad Z(s) = \left(\infty, \infty, \infty, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \frac{L_5 s}{C_5 L_5 s^2 + 1}, \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_4 L_5 R_4 s^4 - R_4 + s^3 (C_4 L_4 L_5 R_4 g_m - C_5 L_4 L_5) + s^2 (-C_4 L_4 R_4 - C_5 L_5 R_4 + L_4 L_5 g_m) + s (-L_4 + L_5 R_4 g_m)}{2 R_4 g_m + s^4 (2 C_4 C_5 L_4 L_5 R_4 g_m + 2 C_4 C_5 L_4 L_5) + s^3 (2 C_4 L_4 L_5 g_m + 2 C_5 L_4 L_5 g_m) + s^2 (2 C_4 L_4 R_4 g_m + 2 C_4 L_4 + 2 C_5 L_5 R_4 g_m + 2 C_5 L_5) + s (2 L_4 g_m + 2 L_5 g_m) + 2}$$

$$\mathbf{9.71 \quad X-INVALID-ORDER-71} \quad Z(s) = \left(\infty, \infty, \infty, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, L_5 s + R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{C_4 C_5 L_4 L_5 R_4 g_m s^4 + R_4 g_m + s^3 (C_4 C_5 L_4 R_4 R_5 g_m - C_4 C_5 L_4 R_4 + C_5 L_4 L_5 g_m) + s^2 (C_4 L_4 R_4 g_m + C_5 L_4 R_5 g_m - C_5 L_4 + C_5 L_5 R_4 g_m) + s (C_5 R_4 R_5 g_m - C_5 R_4 + L_4 g_m)}{2 C_4 C_5 L_4 L_5 g_m s^4 + 2 g_m + s^3 (2 C_4 C_5 L_4 R_4 g_m + 2 C_4 C_5 L_4 R_5 g_m + 2 C_4 C_5 L_4) + s^2 (2 C_4 L_4 g_m + 2 C_5 L_4 g_m + 2 C_5 L_5 g_m) + s (2 C_5 R_4 g_m + 2 C_5 R_5 g_m + 2 C_5)}$$

$$\mathbf{9.72 \quad X-INVALID-ORDER-72} \quad Z(s) = \left(\infty, \infty, \infty, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_4 L_5 R_4 R_5 s^4 - R_4 R_5 + s^3 (C_4 L_4 L_5 R_4 R_5 g_m - C_4 L_4 L_5 R_4 - C_5 L_4 L_5 R_5) + s^2 (-C_4 L_4 R_4 R_5 - C_5 L_5 R_4 R_5 + L_4 L_5 R_5 g_m - L_4 L_5) + s (-L_4 R_5 + L_5 R_4 R_5 g_m - L_5 R_4)}{2 R_4 R_5 g_m + 2 R_5 + s^4 (2 C_4 C_5 L_4 L_5 R_4 R_5 g_m + 2 C_4 C_5 L_4 L_5 R_5) + s^3 (2 C_4 L_4 L_5 R_4 g_m + 2 C_4 L_4 L_5 R_5 g_m + 2 C_4 L_4 L_5 + 2 C_5 L_4 L_5 R_5 g_m) + s^2 (2 C_4 L_4 R_4 R_5 g_m + 2 C_4 L_4 R_5 + 2 C_5 L_5 R_4 R_5 g_m + 2 C_5 L_5 R_5 + 2 L_4 L_5 g_m) + s (2 L_4 R_5 g_m + 2 L_5 R_4 g_m + 2 L_5 R_5 g_m + 2 L_5)}$$

$$\mathbf{9.73 \quad X-INVALID-ORDER-73} \quad Z(s) = \left(\infty, \infty, \infty, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \infty \right)$$

$$H(s) = \frac{R_4 R_5 g_m - R_4 + s^4 (C_4 C_5 L_4 L_5 R_4 R_5 g_m - C_4 C_5 L_4 L_5 R_4) + s^3 (C_4 L_4 L_5 R_4 g_m + C_5 L_4 L_5 R_5 g_m - C_5 L_4 L_5) + s^2 (C_4 L_4 R_4 R_5 g_m - C_4 L_4 R_4 + C_5 L_5 R_4 R_5 g_m - C_5 L_5 R_4 + L_4 L_5 g_m) + s (L_4 R_5 g_m - L_4 + L_5 R_4 g_m)}{2 R_4 g_m + 2 R_5 g_m + s^4 (2 C_4 C_5 L_4 L_5 R_4 g_m + 2 C_4 C_5 L_4 L_5 R_5 g_m + 2 C_4 C_5 L_4 L_5) + s^3 (2 C_4 L_4 L_5 g_m + 2 C_5 L_4 L_5 g_m) + s^2 (2 C_4 L_4 R_4 g_m + 2 C_4 L_4 R_5 g_m + 2 C_4 L_4 + 2 C_5 L_5 R_4 g_m + 2 C_5 L_5 R_5 g_m + 2 C_5 L_5) + s (2 L_4 g_m + 2 L_5 g_m) + 2}$$

$$\mathbf{9.74 \quad X-INVALID-ORDER-74} \quad Z(s) = \left(\infty, \infty, \infty, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \infty \right)$$

$$H(s) = \frac{R_4 R_5 g_m - R_4 + s^4 (C_4 C_5 L_4 L_5 R_4 R_5 g_m - C_4 C_5 L_4 L_5 R_4) + s^3 (-C_4 C_5 L_4 R_4 R_5 + C_5 L_4 L_5 R_5 g_m - C_5 L_4 L_5) + s^2 (C_4 L_4 R_4 R_5 g_m - C_4 L_4 R_4 - C_5 L_4 R_5 + C_5 L_5 R_4 R_5 g_m - C_5 L_5 R_4) + s (-C_5 R_4 R_5 + L_4 R_5 g_m - L_4)}{2 R_4 g_m + 2 R_5 g_m + s^4 (2 C_4 C_5 L_4 L_5 R_4 g_m + 2 C_4 C_5 L_4 L_5 R_5 g_m + 2 C_4 C_5 L_4 L_5) + s^3 (2 C_4 C_5 L_4 R_4 R_5 g_m + 2 C_4 C_5 L_4 R_5 + 2 C_5 L_4 L_5 g_m) + s^2 (2 C_4 L_4 R_4 g_m + 2 C_4 L_4 R_5 g_m + 2 C_4 L_4 + 2 C_5 L_5 R_4 g_m + 2 C_5 L_5 R_5 g_m + 2 C_5 L_5) + s (2 C_5 R_4 R_5 g_m + 2 C_5 R_5 + 2 L_4 R_5 g_m - L_4)}$$

$$\mathbf{9.75 \quad X-INVALID-ORDER-75} \quad Z(s) = \left(\infty, \infty, \infty, \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_4 R_4 s^3 + C_4 L_4 R_4 g_m s^2 - C_5 R_4 s + R_4 g_m}{2 g_m + s^3 (2 C_4 C_5 L_4 R_4 g_m + 2 C_4 C_5 L_4) + s^2 (2 C_4 C_5 R_4 + 2 C_4 L_4 g_m) + s (2 C_4 R_4 g_m + 2 C_5 R_4 g_m + 2 C_5)}$$

$$\mathbf{9.76 \quad X-INVALID-ORDER-76} \quad Z(s) = \left(\infty, \infty, \infty, \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_4 R_4 R_5 s^3 - C_5 R_4 R_5 s + R_4 R_5 g_m - R_4 + s^2 (C_4 L_4 R_4 R_5 g_m - C_4 L_4 R_4)}{2 R_4 g_m + 2 R_5 g_m + s^3 (2 C_4 C_5 L_4 R_4 R_5 g_m + 2 C_4 C_5 L_4 R_5) + s^2 (2 C_4 C_5 R_4 R_5 + 2 C_4 L_4 R_4 g_m + 2 C_4 L_4 R_5 g_m + 2 C_4 L_4) + s (2 C_4 R_4 R_5 g_m + 2 C_4 R_4 + 2 C_5 R_4 R_5 g_m + 2 C_5 R_5) + 2}$$

$$\mathbf{9.77 \quad X-INVALID-ORDER-77} \quad Z(s) = \left(\infty, \infty, \infty, \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{C_4 L_4 R_4 g_m s^2 + R_4 g_m + s^3 (C_4 C_5 L_4 R_4 R_5 g_m - C_4 C_5 L_4 R_4) + s (C_5 R_4 R_5 g_m - C_5 R_4)}{2 g_m + s^3 (2 C_4 C_5 L_4 R_4 g_m + 2 C_4 C_5 L_4 R_5 g_m + 2 C_4 C_5 L_4) + s^2 (2 C_4 C_5 R_4 R_5 g_m + 2 C_4 C_5 R_4 + 2 C_4 L_4 g_m) + s (2 C_4 R_4 g_m + 2 C_5 R_4 g_m + 2 C_5 R_5 g_m + 2 C_5)}$$

$$\mathbf{9.78 \quad X-INVALID-ORDER-78} \quad Z(s) = \left(\infty, \infty, \infty, \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, L_5 s + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{C_4 C_5 L_4 L_5 R_4 g_m s^4 - C_4 C_5 L_4 R_4 s^3 - C_5 R_4 s + R_4 g_m + s^2 (C_4 L_4 R_4 g_m + C_5 L_5 R_4 g_m)}{2 C_4 C_5 L_4 L_5 g_m s^4 + 2 g_m + s^3 (2 C_4 C_5 L_4 R_4 g_m + 2 C_4 C_5 L_4 + 2 C_4 C_5 L_5 R_4 g_m) + s^2 (2 C_4 C_5 R_4 + 2 C_4 L_4 g_m + 2 C_5 L_5 g_m) + s (2 C_4 R_4 g_m + 2 C_5 R_4 g_m + 2 C_5)}$$

$$\mathbf{9.79 \quad X-INVALID-ORDER-79} \quad Z(s) = \left(\infty, \infty, \infty, \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \frac{L_5 s}{C_5 L_5 s^2 + 1}, \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_4 L_5 R_4 s^4 + C_4 L_4 L_5 R_4 g_m s^3 + L_5 R_4 g_m s - R_4 + s^2 (-C_4 L_4 R_4 - C_5 L_5 R_4)}{2 R_4 g_m + s^4 (2 C_4 C_5 L_4 L_5 R_4 g_m + 2 C_4 C_5 L_4 L_5) + s^3 (2 C_4 C_5 L_5 R_4 + 2 C_4 L_4 L_5 g_m) + s^2 (2 C_4 L_4 R_4 g_m + 2 C_4 L_4 + 2 C_4 L_5 R_4 g_m + 2 C_5 L_5 R_4 g_m + 2 C_5 L_5) + s (2 C_4 R_4 + 2 L_5 g_m) + 2}$$

$$\mathbf{9.80 \quad X-INVALID-ORDER-80} \quad Z(s) = \left(\infty, \infty, \infty, \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, L_5 s + R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{C_4 C_5 L_4 L_5 R_4 g_m s^4 + R_4 g_m + s^3 (C_4 C_5 L_4 R_4 R_5 g_m - C_4 C_5 L_4 R_4) + s^2 (C_4 L_4 R_4 g_m + C_5 L_5 R_4 g_m) + s (C_5 R_4 R_5 g_m - C_5 R_4)}{2 C_4 C_5 L_4 L_5 g_m s^4 + 2 g_m + s^3 (2 C_4 C_5 L_4 R_4 g_m + 2 C_4 C_5 L_4 R_5 g_m + 2 C_4 C_5 L_4 + 2 C_4 C_5 L_5 R_4 g_m) + s^2 (2 C_4 C_5 R_4 R_5 g_m + 2 C_4 C_5 R_4 + 2 C_4 L_4 g_m + 2 C_5 L_5 g_m) + s (2 C_4 R_4 g_m + 2 C_5 R_4 g_m + 2 C_5 R_5 g_m + 2 C_5)}$$

$$\mathbf{9.81 \quad X-INVALID-ORDER-81} \quad Z(s) = \left(\infty, \infty, \infty, \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_4 L_5 R_4 R_5 s^4 - R_4 R_5 + s^3 (C_4 L_4 L_5 R_4 R_5 g_m - C_4 L_4 L_5 R_4) + s^2 (-C_4 L_4 R_4 R_5 - C_5 L_5 R_4 R_5) + s (L_5 R_4 R_5 g_m - L_5 R_4)}{2 R_4 R_5 g_m + 2 R_5 + s^4 (2 C_4 C_5 L_4 L_5 R_4 R_5 g_m + 2 C_4 C_5 L_4 L_5 R_5) + s^3 (2 C_4 C_5 L_5 R_4 R_5 + 2 C_4 L_4 L_5 R_4 g_m + 2 C_4 L_4 L_5 R_5 g_m + 2 C_4 L_4 L_5) + s^2 (2 C_4 L_4 R_4 R_5 g_m + 2 C_4 L_4 R_5 + 2 C_4 L_5 R_4 R_5 g_m + 2 C_4 L_5 R_4 + 2 C_5 L_5 R_4 R_5 g_m + 2 C_5 L_5 R_5) + s (2 C_4 R_4 R_5 + 2 L_5 R_4 g_m + 2 L_5 R_5)}$$

9.82 X-INVALID-ORDER-82 $Z(s) = \left(\infty, \infty, \infty, \frac{R_4(C_4L_4s^2+1)}{C_4L_4s^2+C_4R_4s+1}, \frac{C_5L_5R_5s^2+L_5s+R_5}{C_5L_5s^2+1}, \infty \right)$

$$H(s) = \frac{C_4L_4L_5R_4g_ms^3 + L_5R_4g_ms + R_4R_5g_m - R_4 + s^4(C_4C_5L_4L_5R_4R_5g_m - C_4C_5L_4L_5R_4) + s^2(C_4L_4R_4R_5g_m - C_4L_4R_4 + C_5L_5R_4R_5g_m - C_5L_5R_4)}{2R_4g_m + 2R_5g_m + s^4(2C_4C_5L_4L_5R_4g_m + 2C_4C_5L_4L_5R_5g_m + 2C_4C_5L_4L_5) + s^3(2C_4C_5L_5R_4R_5g_m + 2C_4C_5L_5R_4 + 2C_4L_4L_5g_m) + s^2(2C_4L_4R_4g_m + 2C_4L_4R_5g_m + 2C_4L_4 + 2C_4L_5R_4g_m + 2C_5L_5R_4g_m + 2C_5L_5R_5g_m + 2C_5L_5) + s(2C_4R_4R_5g_m + 2C_4R_4 + 2C_5R_5g_m + 2C_5R_5)}$$

9.83 X-INVALID-ORDER-83 $Z(s) = \left(\infty, \infty, \infty, \frac{R_4(C_4L_4s^2+1)}{C_4L_4s^2+C_4R_4s+1}, \frac{R_5(C_5L_5s^2+1)}{C_5L_5s^2+C_5R_5s+1}, \infty \right)$

$$H(s) = \frac{-C_4C_5L_4R_4R_5s^3 - C_5R_4R_5s + R_4R_5g_m - R_4 + s^4(C_4C_5L_4L_5R_4R_5g_m - C_4C_5L_4L_5R_4) + s^2(C_4L_4R_4R_5g_m - C_4L_4R_4 + C_5L_5R_4R_5g_m - C_5L_5R_4)}{2R_4g_m + 2R_5g_m + s^4(2C_4C_5L_4L_5R_4g_m + 2C_4C_5L_4L_5R_5g_m + 2C_4C_5L_4L_5) + s^3(2C_4C_5L_4R_4R_5g_m + 2C_4C_5L_4R_5 + 2C_4C_5L_5R_4R_5g_m + 2C_4C_5L_5R_4) + s^2(2C_4C_5R_4R_5 + 2C_4L_4R_4g_m + 2C_4L_4R_5g_m + 2C_4L_4 + 2C_5L_5R_4g_m + 2C_5L_5R_5g_m + 2C_5L_5) + s(2C_4R_4R_5g_m + 2C_4R_4 + 2C_5R_5g_m + 2C_5R_5)}$$

10 X-INVALID-WZ

10.1 X-INVALID-WZ-1 $Z(s) = \left(\infty, \infty, \infty, R_4 + \frac{1}{C_4s}, \frac{R_5}{C_5R_5s+1}, \infty \right)$

$$H(s) = \frac{-C_4C_5R_4R_5s^2 + R_5g_m + s(C_4R_4R_5g_m - C_4R_4 - C_5R_5) - 1}{2g_m + s^2(2C_4C_5R_4R_5g_m + 2C_4C_5R_5) + s(2C_4R_4g_m + 2C_4R_5g_m + 2C_4 + 2C_5R_5g_m)}$$

Parameters:

Q: $\frac{\sqrt{C_4}\sqrt{C_5}R_4\sqrt{R_5g_m}\sqrt{\frac{g_m}{R_4g_m+1}} + \sqrt{C_4}\sqrt{C_5}\sqrt{R_5}\sqrt{\frac{g_m}{R_4g_m+1}}}{C_4R_4g_m + C_4R_5g_m + C_4 + C_5R_5g_m}$

wo: $\sqrt{\frac{g_m}{C_4C_5R_4R_5g_m + C_4C_5R_5}}$

bandwidth: $\frac{\sqrt{\frac{g_m}{C_4C_5R_4R_5g_m + C_4C_5R_5}}(C_4R_4g_m + C_4R_5g_m + C_4 + C_5R_5g_m)}{\sqrt{C_4}\sqrt{C_5}R_4\sqrt{R_5g_m}\sqrt{\frac{g_m}{R_4g_m+1}} + \sqrt{C_4}\sqrt{C_5}\sqrt{R_5}\sqrt{\frac{g_m}{R_4g_m+1}}}$

K-LP: $\frac{R_5g_m-1}{2g_m}$

K-HP: $-\frac{R_4}{2R_4g_m+2}$

K-BP: $\frac{C_4R_4R_5g_m - C_4R_4 - C_5R_5}{2C_4R_4g_m + 2C_4R_5g_m + 2C_4 + 2C_5R_5g_m}$

Qz: None

Wz: $\frac{\sqrt{-R_5g_m+1}}{\sqrt{C_4}\sqrt{C_5}\sqrt{R_4}\sqrt{R_5}}$

11 X-PolynomialError